
EE 552: Power and Energy Systems

Course Code	EE 552
Course Name	Power and Energy Systems
L-T-P-C	3-1-0-4
Prerequisites	EE 201 Electromechanics or Instructors consent
Intended for	UG
Approval	54th BoA (May 2024)

Course Contents

- **Introduction to power systems and its structure:** Generation, transmission and distribution, substation arrangements. (2 Hours)
- **Representation of power system components:** Generator, Transformers, Transmission lines, line parameters, transmission line design, interference of power lines with communication circuits, line insulators, power cables, per unit system. (10 Hours)
- **Network Analysis** – Concepts of PV, PQ and Slack Bus, Load flow using NR Method, DC load flow, Economic load dispatch, unit commitment (12 Hours)
- **Fault Analysis:** Symmetrical components, symmetrical and unsymmetrical fault calculations, transient analysis (7 Hours)
- **Network Protection:** relaying strategies, circuit breakers, numerical relays (5 Hours)
- **Power System Dynamics:** stability, swing equation, equal area criterion, voltage and frequency control (10 Hours)
- **Energy systems:** Solar photovoltaic, solar thermal, wind energy, energy storage (8 Hours)
- **Smart grid components:** SCADA, Smart sensors, communication (2 Hours)

Laboratory Modules

- NA

Text Books

1. J. J. Grainger and W. D. Stevenson, **Power System Analysis**, Tata McGraw Hill.
2. D. P. Kothari and I. J. Nagrath, **Modern Power System Analysis**, Tata McGraw Hill.
3. A. J. Wood and B. F. Wollenberg, **Power Generation, Operation and Control**, 2nd Edition, Wiley-Blackwell.
4. S. Jha, S. Sen, R. Kumar and D. P. Kothari, **Smart Grid- Fundamental and Applications**, New Age International.

References

1. NA